

of concurrent user accesses without trouble or delay in application access.

- **Privacy and regulatory compliance** - With GDPR (General Data Protection Regulation) and customer personal data security in place for the regulatory restrictions of different countries, CIAM plays an important role in monitoring, reporting and notification alerts.
- **Adaptability** - The CIAM solution should have futuristic architecture to handle easy adoption to any architecture (e.g., cloud adoption, on-premises to cloud synchronisation, etc).

When preparing an integration architecture for an enterprise solution or cloud native adoption, one has to define the technical expectations for a CIAM solution, such as:

- User repository type
- Social ID provider
- Open IdP
- Two-way authentication (certificate, SSL handshake)
- Multi-channel access (Web, mobile, standalone)
- SaaS offering

There are many CIAM frameworks available, like Microsoft AD, IBM Federated IM, Gigya and Janrain, to name a few. Some of them are open source and some proprietary with a commercial licence model for integration.

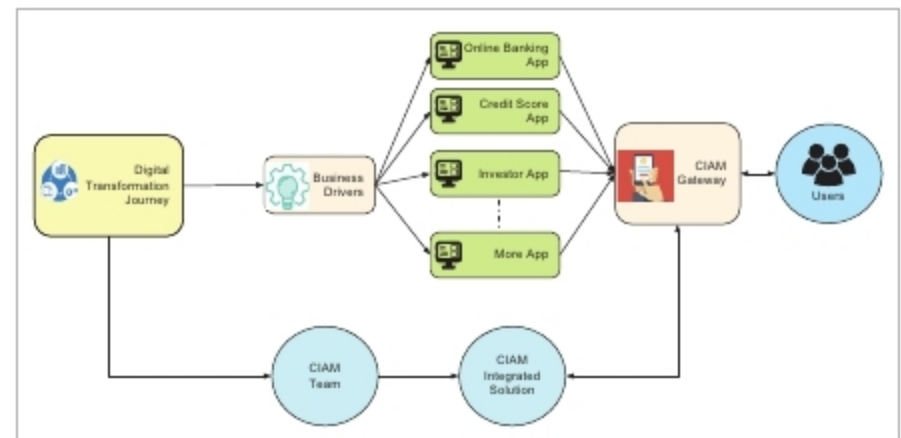


Figure 3: Typical CIAM architecture in a banking solution

Many of them support cloud native adoption, on-premises integration and authorisation using social network accounts. In this series, I will go on to discuss the most popular CIAM frameworks — looking at how to integrate them in the application architecture, while covering their pros and cons in terms of the features supported. **END** 🐧

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```
018-02-11 20:50:58,981 POST Data (www.webscantest.com):
ogin=admin&passwd=pass&subnit_login=login
```

Figure 7: Credentials

```
root@kali:~# driftnet -i eth0 -d /tmp/images
```

Figure 8: Driftnet



Figure 9: Driftnet result 1

```
root@kali:/tmp/images# ls
driftnet-5a75d9886b8b4567.jp
```

Figure 10: Driftnet result 2

The sniffing can be restricted to a particular interface using the `-i` option.

```
driftnet -i <interface> -d <directory
path to save the images>
```

The above command opens a



Figure 11: Driftnet result 3

```
sravya@ubuntu:~$ traceroute google.com
traceroute to google.com (216.58.197.78), 30 hops max, 60 byte packets
 1 192.168.187.145 (192.168.187.145) 0.420 ms
 0.326 ms 0.303 ms
 2 192.168.187.2 (192.168.187.2) 0.463 ms 0.
554 ms 0.482 ms
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
```

Figure 12: Traceroute

window with a view of the images opened by the victim.

As the `-d` option is used, the image is also saved in the given directory.

Testing if someone is sniffing

To find out if someone is sniffing the network, from the victim machine, perform *traceroute* (*tracert* in Windows) to *google.com* (any site hosted outside the local network will work).

We can see that the first hop is 192.168.187.145 (the attacker machine) and then it goes to 192.168.187.2 (the gateway). This type of result shows the clear possibility of network sniffing.

Security measures against attacks

The best security measure against all the above-mentioned data leaks is using full-fledged ssl websites (without any redirects or linking). As the data passing through is encrypted, the sniffer cannot make sense out of it. **END** 🐧

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